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Between cognitive offloading and critical autonomy: a systematic review of the epistemic implications of generative AI in higher education

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Introduction: The rapid integration of generative artificial intelligence (GenAI) into higher education has generated increasing concern regarding its influence on students' epistemic agency, critical thinking, and cognitive autonomy. This systematic review synthesizes current evidence on the epistemic implications of GenAI in higher education, with particular emphasis on the tension between cognitive offloading and critical autonomy.

Methods: Following PRISMA 2020 guidelines, peer-reviewed studies published between 2022 and 2026 were identified through systematic searches in Scopus, the Web of Science Core Collection, and seven education and multidisciplinary databases accessed through EBSCOhost. Eligible studies examined the educational use of generative AI systems such as ChatGPT, Gemini, Claude, and similar large language models within higher education contexts. Data extraction and thematic synthesis were conducted to identify recurring epistemic, cognitive, pedagogical, and ethical patterns across the included studies.

Results: The findings reveal that GenAI simultaneously functions as a cognitive scaffold and a source of epistemic vulnerability. While AI-assisted learning environments may support feedback literacy, metacognitive reflection, conceptual clarification, and self-regulated learning, uncritical reliance on AI-generated outputs was frequently associated across the included studies with cognitive offloading, automation bias, superficial information processing, diminished analytical engagement, and weakened evaluative judgment. The evidence further suggests that the educational consequences of GenAI are mediated primarily by pedagogical design, AI literacy, and assessment practices rather than by the technology itself.

Discussion: Overall, the review supports the need for higher education institutions to promote critical AI literacy and pedagogical frameworks capable of preserving epistemic agency, reflective reasoning, and cognitive autonomy in increasingly AI-mediated learning environments.

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KEYWORDS

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1 Introduction

The rapid expansion of generative artificial intelligence (GenAI), particularly large language models such as ChatGPT, Gemini, Claude, and LLaMA, has transformed higher education environments at an unprecedented pace. Recent evidence suggests that these systems are no longer perceived as occasional technological aids, but rather as normalized components of academic learning ecosystems integrated into writing, information retrieval, summarization, brainstorming, and assessment preparation processes (Cooper et al., 2025; Susena et al., 2026). This accelerated incorporation has shifted educational debates beyond technological adoption toward deeper concerns regarding how students construct, validate, and evaluate knowledge in AI-mediated learning environments.

A growing body of literature has raised concerns regarding the epistemic implications of GenAI use in higher education. Several studies argue that the routine delegation of cognitive tasks to AI systems may foster cognitive offloading, reducing sustained analytical engagement and weakening metacognitive monitoring processes (Raitskaya and Tikhonova, 2025; Schorr et al., 2026). In this context, students may increasingly rely on AI-generated outputs without engaging in adequate verification or critical interrogation of the information produced. Such dynamics are particularly concerning in higher education, where the development of critical thinking and epistemic autonomy represents a central educational objective.

Recent scholarship further suggests that GenAI systems are progressively functioning as epistemic authorities. Because AI-generated responses are frequently presented in fluent, coherent, and authoritative language, students may perceive them as neutral and reliable sources of knowledge even when outputs contain inaccuracies, hallucinations, or hidden biases (Hanna and Yana, 2025; Susena et al., 2026). This tendency contributes to automation bias and epistemic overreliance, phenomena associated with diminished independent judgment and reduced source evaluation practices (Li et al., 2026). Consequently, concerns surrounding AI use extend beyond academic integrity alone and increasingly involve broader questions about the erosion of critical autonomy and responsibility in knowledge validation processes.

At the same time, the literature also recognizes that GenAI may create opportunities for learning enhancement when pedagogically integrated through intentional instructional design. Some studies report that AI-mediated environments can support reflective learning, self-regulated inquiry, metacognitive engagement, and scaffolded reasoning when students are explicitly encouraged to critique, compare, and justify AI-generated outputs (Balart et al., 2026; Fujimoto, 2026). Under these conditions, AI may function not as a substitute for reasoning but as a cognitive support tool capable of facilitating higher-order thinking processes. However, current evidence consistently indicates that these positive outcomes depend heavily on pedagogical framing, assessment design, and AI literacy development rather than on the technology itself (Izquierdo-Condoy et al., 2025).

Parallel concerns have emerged within the fields of information literacy and digital literacy. Traditional frameworks for evaluating information credibility, such as checklist-based

models, have been criticized for their limited ability to address the challenges introduced by AI-generated content and algorithmically mediated information ecosystems (Tomaszewski, 2025). The increasing presence of synthetic authorship, opaque information provenance, and automated content generation requires more sophisticated forms of epistemic evaluation, including source triangulation, lateral reading, and algorithmic literacy (Lin and Tan, 2025). In this context, AI literacy is increasingly conceptualized not only as technical competence but as a critical epistemic capacity involving the evaluation of credibility, transparency, bias, and sociotechnical influence in AI-mediated environments.

Additionally, emerging studies indicate that GenAI may reshape classroom discourse and academic interaction. Research suggests that unguided use of AI-generated content may contribute to homogenized responses, reduced argumentative diversity, and more monologic forms of participation, particularly when students reproduce algorithmic outputs without reinterpretation or critique (Chase and Galvin, 2026; Susena et al., 2026). These concerns are especially relevant in higher education because dialogic interaction, debate, and epistemic disagreement constitute essential mechanisms for disciplinary learning and knowledge construction.

Despite the rapid growth of research in this field, the current literature remains conceptually fragmented. Existing studies frequently examine isolated dimensions of AI use, including academic integrity, technology acceptance, assessment practices, or learning outcomes, while fewer investigations integrate epistemic trust, cognitive offloading, critical thinking, AI literacy, and educational governance within a unified analytical perspective (Raitskaya and Tikhonova, 2025; Susena et al., 2026). Furthermore, the speed of technological evolution and the heterogeneity of conceptual approaches continue to complicate evidence synthesis and pedagogical decision-making.

Given these limitations, there is a growing need for systematic reviews capable of integrating empirical evidence regarding the epistemic implications of GenAI use in higher education. Understanding how AI-mediated educational practices influence knowledge validation, critical autonomy, metacognitive engagement, and responsible reasoning has become essential for the development of pedagogical strategies and institutional policies aligned with the educational mission of universities.

1.1 Research question

How does the use of generative artificial intelligence tools transform epistemic practices in higher education, and what educational responses have been proposed to promote critical and responsible AI use among students?

1.2 General objective

To synthesize the available scientific evidence on the epistemic implications of generative artificial intelligence use in higher education, identifying the associated risks for critical thinking and academic integrity, the opportunities for AI literacy

development, and the educational responses proposed to support responsible and epistemically sound AI integration.

2 Methodology

2.1 Review design

A systematic review with thematic and framework synthesis was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 statement (Page et al., 2021). The review was restricted to original empirical studies employing qualitative, quantitative, or mixed-methods designs. A meta-analysis was not considered appropriate because the included studies examined heterogeneous populations, educational settings, research questions, and outcome constructs and did not provide sufficiently comparable effect estimates.

Thematic synthesis was selected to integrate findings across methodologically diverse empirical studies and to identify recurring patterns concerning epistemic practices, cognitive offloading, critical autonomy, AI literacy, and educational responses. Framework synthesis was subsequently used to organize the synthesized evidence within the epistemological, pedagogical, sociotechnical, and civic-democratic domains established in the review's conceptual framework.

The review question and eligibility criteria were operationalized using the SPIDER framework (Table 1), Sample, Phenomenon of Interest, Design, Evaluation, and Research type, rather than a strict PICOS formulation (Cooke et al., 2012). PICOS was not selected because the review did not evaluate a single intervention against a predefined comparator or seek to estimate a pooled intervention effect. SPIDER was considered

more congruent with the inclusion of qualitative, quantitative, and mixed-methods evidence and with the review's focus on epistemic practices, experiences, risks, and educational responses.

To avoid conflating different levels of inference, all synthesis outputs were classified as: (1) directly reported empirical findings from the included studies; (2) evidence-informed interpretations developed by the review team through thematic or framework synthesis; or (3) normative propositions concerning desirable educational or institutional action. Direct findings and evidence-informed interpretations are reported in the Results section and are explicitly differentiated. Normative propositions are confined to the Discussion and Implications sections.

The review protocol was prospectively registered in PROSPERO under the identifier CRD420261390452 on May 2nd, 2026, before the commencement of screening; direct record: <https://www.crd.york.ac.uk/PROSPERO/view/CRD420261390452>, with the timing and status of the deposit reported transparently.

2.2 Eligibility criteria

2.2.1 Publication status and source type

Eligible publications were original peer-reviewed journal articles available by the search cut-off date either as a version of record or as a publisher-designated early-access, online-first, or article-in-press publication. Early-access and article-in-press records were included only when peer review had been completed and the record had an identifiable journal title, digital object identifier, and publisher-hosted publication page by May 2, 2026.

2.2.2 Inclusion criteria

The publication status of every included article was verified by cross-checking the document type and indexing information provided by the bibliographic database against the DOI metadata and the corresponding publisher page. For early-access or online-first articles, the first online publication date was used for temporal classification. For articles already assigned to a volume and issue, the final bibliographic publication year was used.

Studies were eligible if they reported original empirical data collected from participants in higher education settings, employing qualitative, quantitative, or mixed-methods research designs. Articles were considered eligible for inclusion if they met all of the following criteria:

- Published between January 2022 and May 2026. This temporal boundary was defined by the public availability and rapid educational diffusion of accessible large language model interfaces, most notably ChatGPT, which marked a qualitative shift in the educational presence of generative AI.
- Constituting peer-reviewed empirical articles, whether qualitative, quantitative, or mixed-methods in design, published in indexed academic journals.

TABLE 1 Operationalization of the review question using the SPIDER framework.

SPIDER component	Operational definition
Sample	Undergraduate and postgraduate students, preservice teachers, faculty members, and mixed participant groups situated in higher education institutions.
Phenomenon of Interest	The use or integration of generative artificial intelligence and large language models in academic activities involving the production, mediation, evaluation, validation, or communication of knowledge.
Design	Qualitative studies, analytical cross-sectional studies, mixed-methods studies, quasi-experimental studies, and randomized controlled trials.
Evaluation	Critical thinking, epistemic trust and authority, information and source verification, cognitive offloading, epistemic agency, metacognition, AI literacy, academic integrity, risks, opportunities, and proposed or tested educational responses.
Research type	Original peer-reviewed empirical qualitative, quantitative, and mixed-methods research.

- Situated in higher education contexts, including undergraduate and postgraduate programs, teacher education, health sciences and medical education, language education at the tertiary level, and professional training programs affiliated with higher education institutions.
- Addressing at least one of the following epistemic dimensions in relation to generative AI use: critical thinking or higher-order reasoning; epistemic trust, epistemic authority, or knowledge credibility; source evaluation or information verification; cognitive offloading or epistemic delegation to AI; AI literacy as an epistemic competency; academic integrity understood as an epistemic rather than merely disciplinary concern; or metacognitive engagement with AI-generated content.
- Examining generative artificial intelligence systems, including large language models, ChatGPT and its successive versions, Claude, Gemini, LLaMA, and equivalent tools, in their function of producing, mediating, synthesizing, evaluating, or communicating knowledge in educational contexts.

2.2.3 Exclusion criteria

Records were excluded if they met any of the following conditions:

- Addressed artificial intelligence in education exclusively as a tool for administrative management, student performance prediction, or learning analytics optimization, without connection to epistemic practice, critical reasoning, or knowledge evaluation.
- Were purely technical or computational in nature, without educational application, implication, or discussion of learning outcomes.
- Consisted of non-academic opinion pieces, editorials without methodological grounding, blog posts, or popular science material.
- Focused exclusively on AI-generated content detection mechanisms, plagiarism detection software or AI-writing classifiers, without addressing the epistemic, formative, or pedagogical dimensions of AI use.
- Full text could not be obtained as the version of record or as an author-accepted manuscript corresponding to an identifiable peer-reviewed journal article. A standalone preprint was not considered evidence of an eligible peer-reviewed publication.
- Were conducted exclusively with learner populations below tertiary educational level, unless forming part of a comparative design in which higher education participants constituted a primary analytical group.
- Systematic reviews, scoping reviews, narrative literature reviews, and purely conceptual or theoretical articles were excluded from the synthesis.

Grey literature was not eligible for inclusion. Conference abstracts, non-peer-reviewed proceedings, dissertations, institutional reports, policy documents, unpublished manuscripts, and

preprints without a corresponding peer-reviewed journal record were excluded. This scope restriction was adopted because the review was designed to synthesize peer-reviewed empirical evidence and to apply study-design-specific critical appraisal procedures consistently. The potential for publication and availability bias arising from the exclusion of grey literature is acknowledged in the limitations of the review.

2.3 Search strategy

Systematic electronic searches were conducted on May 2, 2026, in Scopus, accessed through Elsevier; the Web of Science Core Collection, accessed through Clarivate; and ERIC (Education Resource Information Center), Education Source/Education Source Ultimate, Education Research Complete, Teacher Reference Center (TRC), Educational Administration Abstracts, Sociology Source Ultimate, and Academic Search Complete/Ulimate, accessed through the EBSCOhost interface. EBSCOhost is reported as the search platform rather than as a bibliographic database.

The search strategy combined three concept blocks: (1) generative artificial intelligence technologies and large language models; (2) higher education settings and populations; and (3) epistemic practices, knowledge evaluation, cognitive offloading, critical thinking, metacognition, and AI literacy. In Scopus, free-text terms were searched in the title, abstract, and keyword fields. In Web of Science, terms were searched using the Topic field. In the EBSCOhost databases, title, abstract, subject-heading, and thesaurus terms were combined where supported by the individual database.

The complete queries and filters used in each database are available in [Supplementary File S1](#). For each database, the supplement reports the database and platform, coverage searched, complete search string exactly as executed, field codes, Boolean and truncation operators, date and document-type restrictions, search date, and number of records retrieved. Search reporting was revised in accordance with the PRISMA-S extension ([Rethlefsen et al., 2021](#)).

Publisher-designated early-access, online-first, and article-in-press records indexed by the search cut-off date were eligible for screening. No geographic or language restrictions were applied during searching or eligibility assessment.

Grey literature databases, thesis repositories, conference proceedings, preprint servers, and institutional websites were not searched because grey literature was outside the predefined peer-reviewed empirical scope of the review. Reference lists of relevant systematic and scoping reviews identified during searching were nevertheless examined manually to identify potentially eligible primary empirical studies. The reviews themselves were not included in the synthesis.

All retrieved records were exported with complete citation and abstract metadata to EndNote and subsequently imported into Rayyan for deduplication and screening.

2.4 Study selection

Following deduplication of all retrieved records, the screening process was conducted using the Rayyan Professional systematic

review platform (Valizadeh et al., 2022) to facilitate blinded and independent study selection. Two independent reviewers screened titles and abstracts against the pre-specified eligibility criteria. Decisions were recorded as include, exclude, or uncertain. Disagreements were resolved through structured discussion between the reviewers; cases that remained unresolved after discussion were adjudicated by a third reviewer. Inter-rater reliability at the abstract screening stage was quantified using Cohen's kappa coefficient (κ), with a minimum threshold of $\kappa \geq 0.70$ established *a priori* as indicative of substantial agreement. Full-text copies of all records passing the abstract screening stage were retrieved and subjected to independent eligibility assessment by the same two reviewers applying the complete set of inclusion and exclusion criteria. For each record excluded at the full-text stage, the specific reason for exclusion was documented.

The automated assessment generated an eligibility recommendation and a criterion-specific justification for each evaluated record. Records containing insufficient, incomplete, or ambiguous information were retained for human screening rather than excluded automatically. Relevance ranking, keyword occurrence, or the absence of an individual search term was not used as the sole basis for exclusion. The purpose of the procedure was to apply the prespecified eligibility criteria consistently across a large corpus while prioritizing uncertain and potentially relevant records for independent reviewer assessment.

2.5 Data extraction

A structured data extraction instrument was developed and piloted on a random sample of ten studies prior to full application. Following piloting, the instrument was refined to ensure clarity and consistency of variable definitions. Extraction was performed independently by two reviewers; discrepancies were resolved by discussion and, where necessary, by third-reviewer adjudication. The following dimensions and variables were extracted for each included study.

2.6 Quality appraisal

Methodological quality and risk of bias were independently assessed by two reviewers using Joanna Briggs Institute (JBI) critical appraisal tools (Aromataris and Munn, 2020), appropriate to each empirical study design. Depending on methodological classification, the following instruments were applied: JBI Checklist for Qualitative Research, Analytical Cross-Sectional Studies, Randomized Controlled Trials, Quasi-Experimental Studies, Cohort Studies, and Mixed-Methods Studies. Disagreements were resolved through consensus discussion. Studies were not excluded based on methodological quality alone, but appraisal outcomes informed the interpretive weighting of evidence during synthesis. Mixed-methods studies were appraised by evaluating their qualitative and quantitative components separately using the corresponding JBI design-specific tools.

2.7 Synthesis approach

The evidence synthesis was organized across three sequential and complementary levels.

At the first level, a descriptive mapping documented the formal characteristics of the included corpus: temporal distribution of publications; geographic and institutional provenance of studies; educational levels and disciplinary fields; study designs; AI technologies examined and their educational functions; and frequency and distribution of conceptual and theoretical frameworks. Descriptive statistics were computed where appropriate, and results were presented in summary tables.

At the second level, thematic synthesis was conducted following the three-stage procedure described by Thomas and Harden (Thomas and Harden, 2008): (i) line-by-line coding of the results and findings sections of all included primary studies; (ii) construction of descriptive themes through the grouping of semantically related codes; and (iii) generation of analytical themes through an interpretive process that moved beyond the descriptive content of individual studies to produce new conceptual understandings relevant to the review questions. Coding and thematic construction were performed by two researchers and subjected to iterative team discussion until analytical consensus was achieved.

At the third level, framework synthesis (Carroll et al., 2011) was applied to map the synthesized evidence onto the four-domain conceptual architecture, epistemological, pedagogical, sociotechnical, and civic-democratic, developed in the conceptual background of the review. The fit between empirical findings and framework components was systematically assessed, and the framework was refined where the evidence warranted modification or extension of its original structure. The resulting integrative model, designated *Hybrid Epistemic Engagement Practices in AI-Mediated Higher Education* (HEEP-HE), constitutes the principal theoretical contribution of the review.

Given the conceptual, methodological, and outcome heterogeneity of the included studies, a formal statistical meta-analysis was not considered appropriate. Instead, narrative synthesis served as the primary integration strategy and was conducted in accordance with the Synthesis Without Meta-analysis (SWiM) reporting guideline (Campbell et al., 2020). The synthesis combined descriptive mapping, thematic synthesis, and framework synthesis to ensure methodological transparency, systematic organization of evidence, and interpretive integration of qualitative and non-comparable findings.

To preserve transparency regarding the epistemic status of the review's claims, coded statements were differentiated throughout synthesis as direct empirical findings, evidence-informed review-level interpretations, or normative propositions. Direct empirical findings reflected results explicitly reported by one or more included studies. Evidence-informed interpretations were generated by the review team through thematic or framework synthesis and are described as interpretive rather than directly observed results. Normative propositions concerning what institutions, educators, or policymakers should do were not treated as empirical findings and were reserved for the Discussion.

3 Results

3.1 Selection process

Inter-reviewer agreement across title and abstract and full-text screening was high, with an overall Cohen's kappa coefficient of $\kappa = 0.85$.

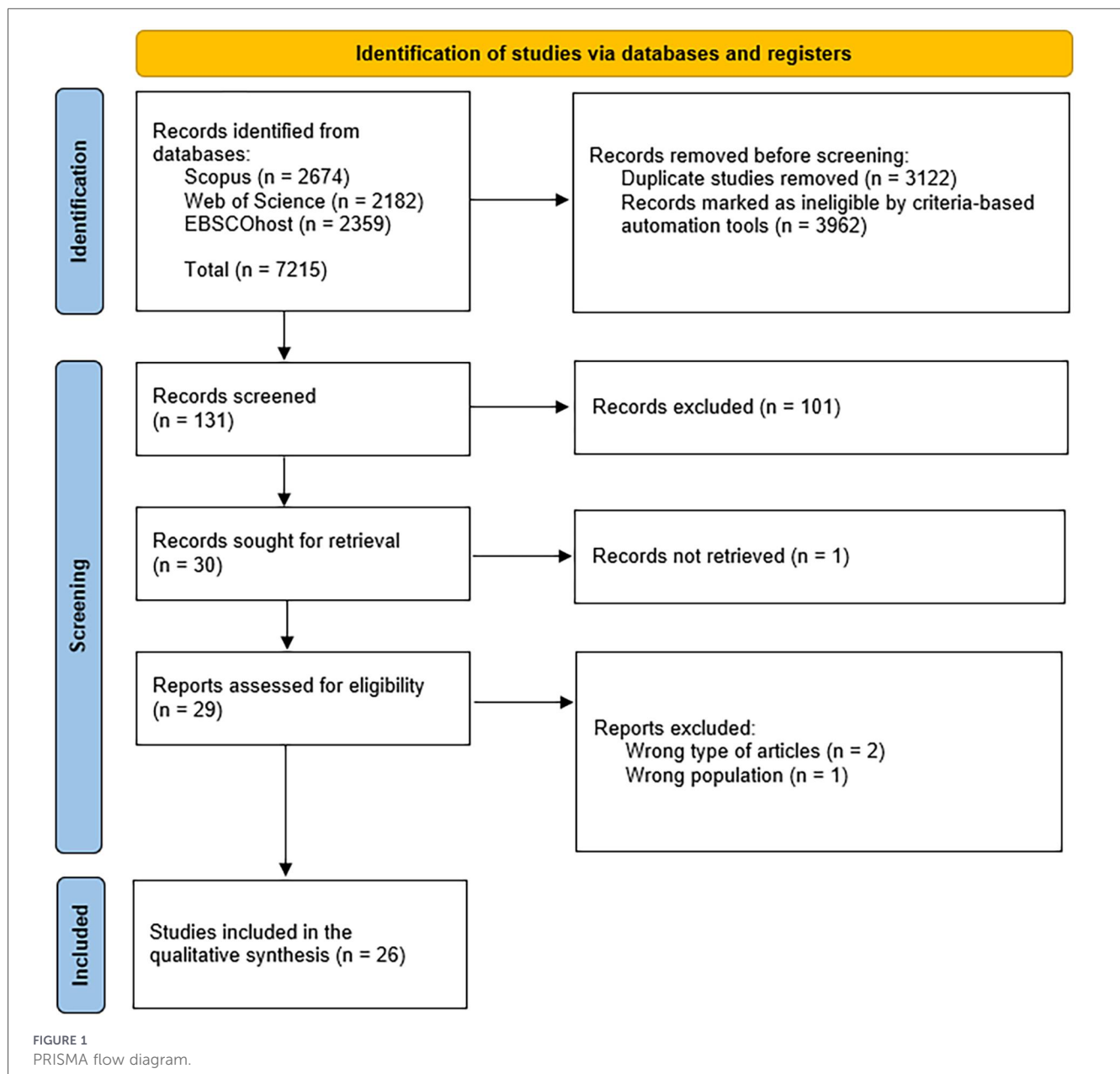
Figure 1 presents the study selection process in accordance with the PRISMA 2020 statement. A total of 7,215 records were identified through database searching: 2,674 from Scopus, 2,182 from the Web of Science Core Collection, and 2,359 from the databases searched through the EBSCOhost platform.

Before title and abstract screening, 3,122 duplicate records were removed. The remaining 4,093 unique records underwent the prespecified criteria-based automated eligibility assessment conducted in Rayyan. This procedure marked 3,962 records as ineligible because they clearly met one or more predefined

exclusion criteria. Records with incomplete or ambiguous eligibility information were retained for human assessment. Consequently, 131 records proceeded to blinded and independent title and abstract screening.

During title and abstract screening, 101 records were excluded because they did not meet the eligibility criteria. Full texts were sought for the remaining 30 records; however, one article could not be retrieved. Therefore, 29 full-text articles were independently assessed for eligibility. Three articles were excluded at this stage: two because of an ineligible publication type (Peters and Angelov, 2025, Vendrell and Johnston, 2026) and one because its study population was outside the scope of the review (Uğraş et al., 2025).

Ultimately, 26 studies (Al-atrash et al., 2026; AlBannai, 2025; Alhur et al., 2025; Alkaabi et al., 2025; Allaire et al., 2026; Andewi et al., 2025; Athanassopoulos et al., 2026; Ba et al., 2025; Borzanović et al., 2026; Brummernhenrich et al., 2025;



Cevikbas and Kaiser 2026; Erlangga et al., 2026; Härkki et al., 2025; Hong et al., 2025; Hua and Cunningham, 2026; Huang et al., 2024; Li et al., 2025; Pachava et al., 2026; Pakdel et al., 2025; Pandey et al., 2026; Thi, 2024; Urban et al., 2025; Vinyard and Roosa, 2025; Wang and Liu, 2025; Waziana et al., 2025; Zhang et al., 2024) met all eligibility criteria and were included in the final qualitative synthesis.

3.2 Descriptive synthesis results

Although several included studies initially focused on educational performance domains such as writing assistance, formative assessment, or learning efficiency, they were retained in the synthesis when their findings directly addressed epistemic processes related to knowledge validation, cognitive delegation, authorship, information trustworthiness, evaluative judgment, or learner autonomy in AI-mediated environments. Eligibility decisions therefore prioritized epistemic implications rather than disciplinary context alone.

A total of 26 studies were included in the final synthesis. The included evidence demonstrated a marked temporal concentration in the period 2025–2026, indicating that research concerning generative artificial intelligence and epistemic practices in higher education had rapidly expanded during the most recent years of the literature. Specifically, three studies had been published in 2024, fourteen in 2025, and nine in 2026.

No eligible studies were published in 2022 or 2023. Although the eligibility window began in 2022 to capture the earliest literature following the public diffusion of conversational generative AI, the first publications meeting all three core eligibility requirements, an empirical design, a higher education setting, and an explicit epistemic focus, appeared in 2024. Records retrieved from 2022 to 2023 did not satisfy the complete set of eligibility criteria. This temporal lag indicates that the empirical evidence base remains recent and still developing.

The geographical distribution of the corpus revealed a predominance of studies conducted within Global South contexts. Eighteen studies originated from institutions located in the Global South, seven were conducted in the Global North, and one study incorporated a mixed North–South collaborative perspective. The evidence base was therefore characterized by strong representation from Asian, Middle Eastern, and other non-Western higher education systems, suggesting that concerns surrounding AI-mediated knowledge validation had emerged as a global educational phenomenon rather than an exclusively Western debate.

Methodologically, the corpus demonstrated substantial heterogeneity, including 11 qualitative studies, 7 mixed-methods studies, 4 analytical cross-sectional studies, 2 quasi-experimental studies, and 2 randomized controlled trials. The reviewed evidence, therefore, reflected an emergent field characterized simultaneously by exploratory inquiry, empirical measurement, and intervention-oriented experimentation.

All included studies were situated within higher education settings, although important disciplinary differences were identified. The reviewed evidence encompassed medical education, STEM disciplines, engineering, teacher education, humanities, social sciences, English as a Foreign Language

instruction, information science, and technology-enhanced learning environments. Such disciplinary diversity indicated that epistemic transformations associated with generative AI were not restricted to a single academic domain, but rather affected multiple forms of disciplinary knowledge production and validation.

ChatGPT emerged as the dominant technological artifact across the corpus and was explicitly examined in all included studies. Additional generative AI systems, including Gemini, Claude, Copilot, NotebookLM, Perplexity, Quillbot, Grammarly, InquiryGPT, and AlphaFold-integrated systems, were also identified. Nevertheless, the evidence overwhelmingly focused on large language model-based conversational systems functioning as mediators of learning, writing, reasoning, and information synthesis.

Across the reviewed literature, generative AI technologies were most frequently employed for writing assistance, ideation, feedback generation, information retrieval, conceptual explanation, problem solving, and research support. These functions positioned AI systems not merely as productivity tools, but as epistemic intermediaries capable of influencing how knowledge was accessed, interpreted, validated, and communicated.

The synthesis further demonstrated that the principal epistemic practices addressed in the literature included critical thinking, information validation, authorship negotiation, epistemic agency, metacognition, argumentation, trust calibration, and evaluation of evidence. Consequently, the reviewed studies collectively suggested that generative AI had begun to reshape the epistemic architecture of higher education by modifying relationships between learners, knowledge authorities, and processes of truth verification.

Across the included studies, recurrent risks concerned overreliance, inaccurate or unverifiable outputs, cognitive offloading, weakened evaluative judgment, and ambiguous authorship. Reported opportunities included feedback, conceptual scaffolding, accessibility, reflective learning, and self-regulation. The educational responses proposed or tested across the corpus focused primarily on AI literacy, verification practices, assessment redesign, faculty development, and institutionally governed integration. These patterns are examined in detail in the thematic synthesis. (Table 2)

3.3 Risk of bias appraisal

As shown in Figure 2, the methodological quality assessment using the JBI critical appraisal tools demonstrated generally favorable methodological appraisal profiles. The included studies received predominantly positive item-level assessments across the applicable JBI criteria. The proportion of affirmative responses ranged from 80.0% to 100.0%; these percentages are reported descriptively and were not interpreted as validated summary risk-of-bias categories. Qualitative research represented the largest proportion of included studies ($n = 11$), followed by mixed-methods studies ($n = 7$), analytical cross-sectional studies ($n = 4$), quasi-experimental studies ($n = 2$), and randomized controlled trials ($n = 2$).

Most studies fulfilled key appraisal criteria related to the clarity of objectives, appropriateness of methodology, coherence between

TABLE 2 Comprehensive descriptive mapping of included studies.

Study	Year	Global Context	Methodological Approach	Educational/ Disciplinary Context	AI Technology Examined	Primary Educational Function of AI	Main Epistemic Practices Addressed	Principal Risks Identified	Principal Opportunities Identified	Main Educational Responses/ Recommendations
Al-atrash et al. 2026	2026	Global South	QLR	Higher education - multidisciplinary	ChatGPT, generative AI	Writing assistant, information search, idea generation	Academic integrity, critical source evaluation, citation, paraphrasing	Plagiarism, academic dishonesty, dependency, biases, hallucinations	Learning personalization, immediate feedback, efficiency	Institutional policies, AI literacy, assessment redesign, teacher training
AlBannai 2025	2025	Global South	QLR	Higher education	ChatGPT, generative AI	Learning support, writing, comprehension	Self-regulated learning, critical thinking, evaluation	Over-reliance, academic integrity, accuracy	Learning improvement, efficiency, accessibility	Policies, training, curricular integration
Alhur et al. 2025	2025	Global South	QLR	Health sciences/ higher education	ChatGPT	Academic assistance, search, writing	Critical thinking, academic integrity, information validation	Hallucinations, dependency, plagiarism	Efficiency, feedback, personalization	AI literacy training, institutional policies
Alkaabi et al. 2025	2025	Global South	QLR	Higher education	ChatGPT	Learning, writing, research	Critical thinking, self-regulation, source evaluation	Over-reliance, accuracy, ethics	Learning support, efficiency	Pedagogical integration, training
Allaire et al. 2026	2026	Global North	MMS	Higher education	Generative AI, ChatGPT	Knowledge construction, academic writing	Critical thinking, epistemic agency, evaluation	Erosion of epistemic skills, hallucinations	Learning scaffolding, feedback	Critical AI pedagogy, epistemic literacy
Andewi et al. 2025	2025	Global South	MMS	Higher education - English language	ChatGPT	Writing assistance, grammar, vocabulary	Autonomous learning, feedback	Dependency, accuracy	Skills improvement, motivation	Curricular integration, training
Athanassopoulos et al. 2026	2026	Global North	MMS	Higher education	ChatGPT	Writing assistance, brainstorming, revision	Reflection, critical thinking, authorship	Over-reliance, originality, integrity	Productivity, scaffolding	Design of AI-resistant assignments
Ba et al. 2025	2025	Global South	QES	Higher education	ChatGPT, generative AI	Learning support	Self-regulation, critical thinking	Over-reliance, accuracy	Personalization, efficiency	Pedagogical integration

(Continued)

TABLE 2 Continued

Study	Year	Global Context	Methodological Approach	Educational/ Disciplinary Context	AI Technology Examined	Primary Educational Function of AI	Main Epistemic Practices Addressed	Principal Risks Identified	Principal Opportunities Identified	Main Educational Responses/ Recommendations
Borzanović et al. 2026	2026	Global North	MMS	Higher education	ChatGPT	Research, writing, learning	Critical thinking, source evaluation	Academic integrity, hallucinations	Study support, feedback	Institutional policies, training
Brummernhenrich et al. 2025	2025	Global North	RCT	Higher education	ChatGPT	Educational dialogue, feedback	Critical thinking, evaluation, epistemic trust	Hallucinations, overconfidence	Personalization, immediate feedback	Dialogic design, epistemic literacy
Cevikbas and Kaiser 2026	2026	Mixed	QLR	Higher mathematics education	ChatGPT	Problem solving, concept explanation	Mathematical reasoning, validation	Mathematical errors, dependency	Personalization, scaffolding	Specific didactic design, validation
Erlangga et al. 2026	2026	Global South	CSS	Higher education	ChatGPT	Academic assistance	Learning, self-regulation	Dependency, ethics	Efficiency, accessibility	Curricular integration
Härkki et al. 2025	2025	Global North	QLR	Higher education	ChatGPT, generative AI	Knowledge co-construction	Critical thinking, epistemic agency, collaboration	Skill erosion, hallucinations	Collaboration support, scaffolding	Critical AI pedagogy
Hong et al. 2025	2025	Global South	MMS	Higher education	ChatGPT	Learning support, writing	Critical thinking, creativity	Over-reliance, cognitive atrophy	Skills improvement	Structured pedagogical design
Hua and Cunningham 2026	2026	Global South	QLR	Higher education	ChatGPT, generative AI	Study support	Self-regulation, critical thinking	Dependency, integrity	Personalization	Pedagogical integration
(Huang et al. 2024)	2024	Global South	CSS	Higher education	ChatGPT, generative AI	Learning, writing	Self-regulated learning, evaluation	Dependency, accuracy	Efficiency, motivation	Curricular integration
Li et al. 2025	2025	Global South	RCT	Higher STEM education	InquiryGPT (ChatGPT-based with POEE framework)	Inquiry scaffolding, feedback, reflective evaluation	Critical thinking, problem solving, creativity, knowledge construction	Over-reliance, incorrect responses	Inquiry-based learning, improvement of higher-order thinking	POEE inquiry-guided assistant, constraints to prevent dependency
Pachava et al. 2026	2026	Global South	QLR	Higher education - professional (Engineering, Pharmacy, Management, Law)	ChatGPT, Gemini, Copilot, Claude	Learning substitutes, efficiency-seeking	Critical thinking, academic integrity, epistemic agency, ethics	Over-reliance, cognitive offloading, peer-pressure normalization,	Accessibility, efficiency, feedback	Institutional reforms, ethical AI literacy, teacher training, clear policies

(Continued)

TABLE 2 Continued

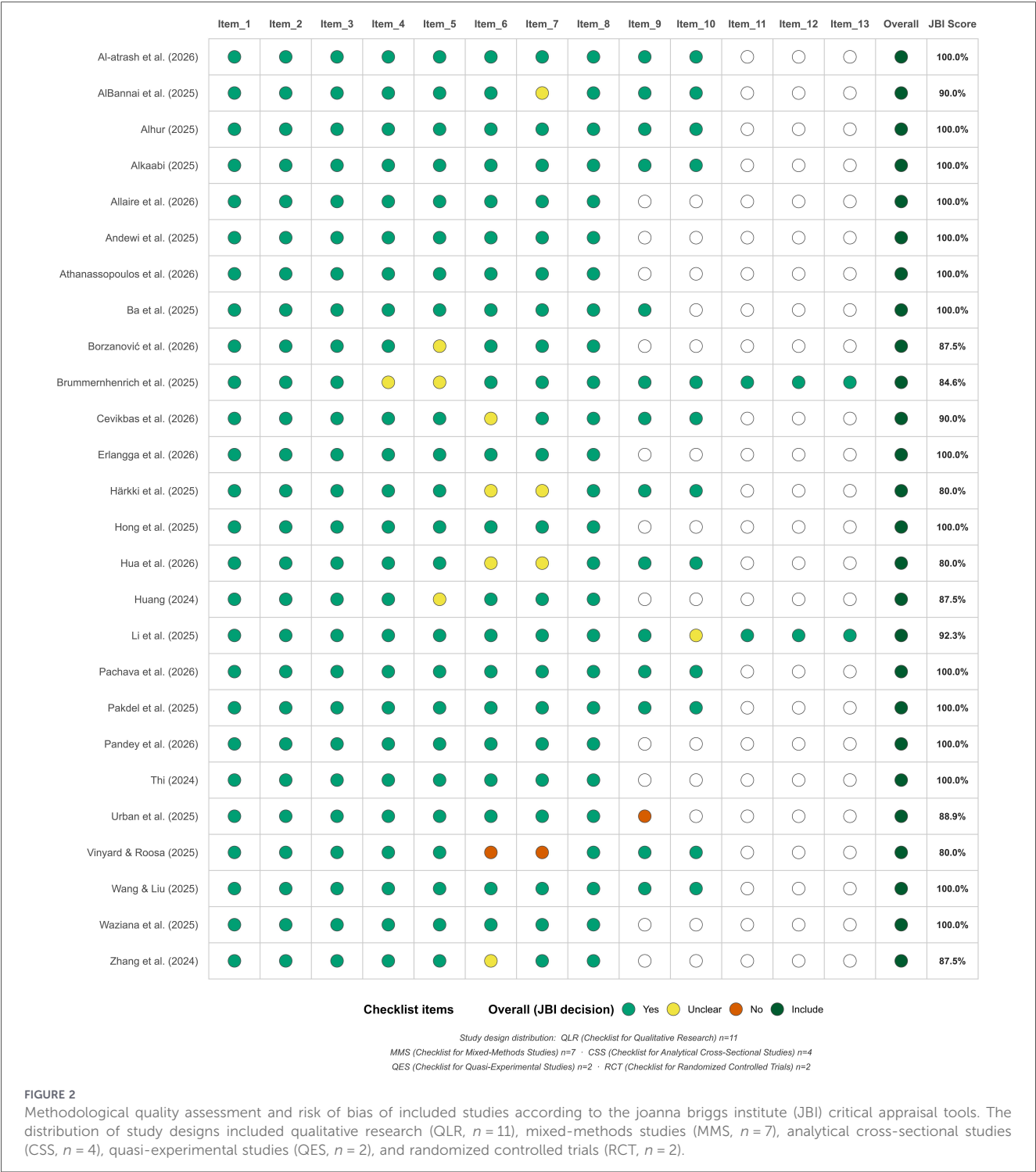
Study	Year	Global Context	Methodological Approach	Educational/ Disciplinary Context	AI Technology Examined	Primary Educational Function of AI	Main Epistemic Practices Addressed	Principal Risks Identified	Principal Opportunities Identified	Main Educational Responses/ Recommendations
Pakdel et al. 2025	2025	Global South	QLR	Higher medical education - L2 academic English	ChatGPT, Gemini, Perplexity, AI Talk (free versions)	Writing assistance, academic vocabulary, grammar, ideation	Linguistic skills, critical thinking, self-regulation	Over-reliance lack of contextual understanding, inaccuracies, risks to creativity	Vocabulary improvement, linguistic refinement, creative writing	Curricula with critical AI evaluation, higher-order thinking assignments
Pandey et al. 2026	2026	Global South	CSS	Higher education	Generative AI (ChatGPT, Gemini, AlphaFold)	Epistemic authority, knowledge source	Epistemic trust in AI: cognitive, contextual, dependability, transparency, performance expectancy, autonomy	Over-reliance on AI, epistemic paralysis, biases	Calibrated trust, critical evaluation	Epistemic literacy, education in AI evaluation
Thi 2024	2024	Global South	MMS	Higher education - EFL academic English	ChatGPT, Gemini, Copilot, Quillbot, Grammarly	Writing assistance, grammar, ideation, paraphrasing	Autonomous learning, critical thinking, integrity	Over-reliance, risk to creativity and critical thinking, misinformation	Linguistic refinement, ideation, feedback integration	Teacher support, AI use training, curricular integration
Urban et al. 2025	2025	Global North	QES	Higher education	ChatGPT-4o	Information source, epistemic authority	Epistemic beliefs, metacognitive accuracy, information evaluation, epistemic vigilance	Over-reliance, integration of incorrect information, miscalibrated trust	Improvement in correct information integration, scaffolding	Education in epistemic literacy, critical evaluation, metacognitive interventions
Vinyard and Roosa 2025	2025	Global North	QLR	Higher education - library research	ChatGPT, Grammarly, Copilot, Bard/ Gemini	Research, source search, summaries, ideation	Critical source evaluation, academic integrity, prompt engineering	False citations (hallucinations), accuracy, plagiarism, dependency	Search efficiency, brainstorming, refinement	Librarians as AI literacy guides, integration with databases

(Continued)

TABLE 2 Continued

Study	Year	Global Context	Methodological Approach	Educational/ Disciplinary Context	AI Technology Examined	Primary Educational Function of AI	Main Epistemic Practices Addressed	Principal Risks Identified	Principal Opportunities Identified	Main Educational Responses/ Recommendations
Wang and Liu 2025	2025	Global South	QLR	Higher education - educational technology	ChatGPT premium, NotebookLM, Gemini, Perplexity, Gamma	Question refinement, academic reasoning, study	Critical thinking, self-regulation, epistemic literacy, inquiry	Accuracy, over-reliance, biases, digital divide	Strategic workflows, POE inquiry, institutional scaffolding	POE model (Predict-Observe-Explain), teacher training, AI literacy
Waziana et al. 2025	2025	Global South	MMS	Higher EFL education - Information Systems	ChatGPT	Writing assistance, ideation, feedback	Critical, creative, and reflective thinking; metacognition	Cognitive offloading, skill atrophy, misinformation, biases, cognitive overload	Cognitive scaffolding, ideation, argument refinement	Critical AI literacy, process evaluation, prompt engineering
Zhang et al. 2024	2024	Global South	CSS	Higher education	ChatGPT	Academic assistance, emotional support, search	Epistemic trust, epistemic reliance, evaluation	cognitive reliance on generative AI, laziness, plagiarism, misinformation, reduction of creativity/critical/independent thinking	Academic support, efficiency	Reduce academic stress, AI literacy, mitigate epistemic reliance

QLR, qualitative research; MMS, mixed-methods study; CSS, analytical cross-sectional study; QES, quasi-experimental study; RCT, randomized controlled trial.



methods and outcomes, and adequacy of statistical or thematic analyses. Several studies obtained perfect scores (100%), reflecting strong methodological consistency and transparent reporting practices. However, some studies presented minor methodological concerns, primarily associated with unclear reporting of confounding factors, incomplete description of follow-up procedures, limited reflexivity in qualitative designs, or insufficient detail regarding participant blinding and allocation concealment in experimental studies.

A small number of studies demonstrated moderate methodological limitations, reflected by “Unclear” or “No” ratings in specific items. The most recurrent issues included inadequate reporting of strategies to manage confounding variables, limited transparency in participant recruitment procedures, and incomplete description of analytical rigor. Despite these limitations, all studies were retained in the review because their overall methodological quality remained acceptable and their findings were considered relevant to the objectives of the synthesis.

3.4 Thematic synthesis

3.4.1 Coding strategy

The thematic synthesis was conducted through an interpretive coding of the extracted fields related to key findings, epistemic practices, risks, opportunities, and educational responses. Initial codes were generated from repeated semantic units across the included studies and were subsequently grouped into descriptive themes. These descriptive themes were then interpreted at a higher level of abstraction to generate analytical themes concerning how generative AI had transformed knowledge validation practices in higher education.

3.4.2 Descriptive themes

As summarized in Table 3, the thematic synthesis identified six interrelated descriptive themes that collectively explained how generative AI had transformed epistemic practices, knowledge validation processes, and educational responses within higher education contexts.

3.4.2.1 Theme 1. Reconfiguration of epistemic authority

As presented in Table 3, the first thematic cluster grouped codes associated with AI-supported explanation, academic assistance, co-construction of knowledge, and epistemic mediation.

Across the included studies, generative AI was not described merely as a technical instrument for productivity. It was repeatedly positioned as a mediator of academic reasoning, writing, explanation, information retrieval, feedback, and knowledge construction. In this role, AI systems were used to support drafting, ideation, conceptual explanation, problem solving, inquiry, source search, and reflective dialogue. This pattern suggested that generative AI had entered higher education as an actor that shaped how students accessed, organized, interpreted, and communicated knowledge.

The main epistemic implication was that authority was partially redistributed from traditional academic sources, instructors, libraries, and disciplinary procedures toward conversational AI systems. This redistribution was not absolute, but it introduced a new epistemic dependency: learners increasingly interacted with AI systems as if they were sources of explanation, validation, and academic orientation.

3.4.2.2 Theme 2. Fragility of knowledge validation and source evaluation

As shown in Table 3, a second thematic cluster emerged around practices of verification, triangulation, source evaluation, and evidence checking.

A second descriptive theme concerned the difficulty of distinguishing plausible AI-generated outputs from reliable knowledge. Several studies identified hallucinations, inaccurate responses, false citations, incomplete contextual understanding, and misinformation as recurrent risks. These risks were especially relevant because AI-generated responses were often coherent, fluent, and rhetorically persuasive, even when they lacked evidentiary grounding.

The thematic pattern indicated that the central challenge was not simply whether AI produced errors, but whether students possessed the epistemic capacity to detect, interpret, and correct those errors. Source triangulation, external verification, citation checking, and comparison with disciplinary evidence emerged as necessary practices for sustaining knowledge validity in AI-mediated environments.

3.4.2.3 Theme 3. Transformation of academic integrity into epistemic integrity

According to the coding structure summarized in Table 3, integrity-related findings extended beyond conventional plagiarism concerns and converged around broader issues of epistemic accountability.

The reviewed studies consistently addressed plagiarism, authorship, paraphrasing, and ethical use. However, the thematic synthesis indicated that the problem exceeded traditional academic misconduct. Integrity was increasingly framed as a broader question of epistemic responsibility: who was accountable for claims, interpretations, sources, and conclusions when AI contributed to the production of academic work?

This theme suggested that the debate had moved from detecting whether AI had been used toward determining how AI had been used, whether its contribution had been disclosed, whether claims had been verified, and whether students retained responsibility for the final epistemic product. Thus, academic integrity was transformed into epistemic integrity, understood as responsible participation in hybrid human–AI knowledge production.

3.4.2.4 Theme 4. Cognitive offloading and erosion of learner agency

As detailed in Table 3, codes related to overreliance, dependency, scaffolding, and self-regulated learning converged into a broader tension between cognitive support and erosion of learner agency.

The corpus showed a persistent tension between AI as scaffold and AI as substitute. On the one hand, AI was associated with personalization, immediate feedback, improved writing, accessibility, motivation, inquiry support, and conceptual scaffolding. On the other hand, studies warned that overreliance could produce cognitive offloading, reduced effort, dependency, loss of creativity, weakening of critical thinking, and erosion of independent judgment.

This theme indicated that the educational value of generative AI depended less on the mere availability of the technology and more on the pedagogical conditions under which it was used. When AI use was guided, reflective, and verification-oriented, it could support higher-order learning. When it was unstructured, convenience-driven, or assessment-driven, it could displace the cognitive work required for epistemic development.

3.4.2.5 Theme 5. Generative AI as scaffold for higher-order thinking

As illustrated in Table 3, several studies associated structured pedagogical integration of AI with opportunities for enhancing higher-order cognitive processes.

Several studies reported opportunities for improving critical thinking, creativity, reflective thinking, inquiry-based learning, argument refinement, and self-regulated learning. These gains

TABLE 3 Coding matrix for thematic synthesis.

Analytical Theme	First-order Codes	Descriptive Theme	Main Risk Pattern	Main Opportunity Pattern	Educational Response Pattern	Representative Studies
Reconfiguration of epistemic authority	AI as source of knowledge; AI as assistant; AI as pseudo-expert; AI as interlocutor; AI as co-constructor of knowledge	Generative AI was positioned as an epistemic mediator in academic tasks	Overtrust, miscalibrated confidence, dependence on AI outputs	Immediate access to explanations, feedback, and academic support	Epistemic literacy, AI literacy, critical evaluation of AI-generated content	Brummernhenrich et al.; Pandey et al.; Urban et al.; Zhang et al.; Härkki et al.
Fragility of knowledge validation and source evaluation	Source evaluation; evidence checking; citation; triangulation; verification; hallucinated references	Students and faculty were required to verify AI-generated information against external sources	Hallucinations, misinformation, inaccurate responses, false citations	Faster information retrieval, synthesis, and refinement of ideas	Verification protocols, information literacy, librarian-guided AI literacy, structured validation practices	Al-atrash et al.; Alhur et al.; Borzanović et al.; Vinyard and Roosa; Urban et al.; Cevikbas and Kaiser.
Transformation of academic integrity into epistemic integrity	Plagiarism; authorship; paraphrasing; transparency; responsibility; ethical use	Integrity concerns shifted from misconduct detection to responsibility for AI-mediated knowledge production	Plagiarism, ambiguous authorship, dishonesty, ethical uncertainty	Support for drafting, revision, ideation, and academic writing	Institutional policies, ethical AI guidelines, assessment redesign, transparent use declarations	Al-atrash et al.; Athanassopoulos et al.; Pachava et al.; Thi; Li et al.
Cognitive offloading and erosion of learner agency	Overreliance; cognitive offloading; reduced effort; atrophy of skills; loss of agency	AI use created tensions between support and substitution of learning processes	Dependency, cognitive atrophy, reduced independent thinking, erosion of epistemic skills	Cognitive scaffolding, self-regulated learning, motivation, accessibility	Metacognitive instruction, guided use, task designs that preserve learner agency	Allaire et al.; Hong et al.; Waziana et al.; Zhang et al.; Pakdel et al.; Pachava et al.
Generative AI as scaffold for higher-order thinking	Inquiry scaffolding; feedback; argument refinement; creativity; problem solving; conceptual explanation	AI was used to support reasoning, reflection, and inquiry when pedagogically structured	Incorrect answers, superficial reasoning, uncritical acceptance	Critical thinking, creativity, reflection, problem solving, learning personalization	POE/POEE models, guided inquiry systems, critical prompting, higher-order assessment tasks	Li et al.; Wang and Liu; Waziana et al.; Cevikbas et al.; Härkki et al.; Brummernhenrich et al.
Institutional transition from prohibition to governed integration	Policy development; curriculum integration; faculty training; AI literacy; assessment reform	Institutions were urged to move from reactive restriction to pedagogically governed integration	Policy gaps, inconsistent guidance, peer normalization of problematic use	Responsible AI adoption, faculty development, curriculum innovation	Institutional governance, teacher training, AI competency frameworks, process-based assessment	AlBannai et al.; Alkaabi; Erlangga et al.; Wang and Liu; Pachava et al.; Vinyard and Roosa.

were most consistently associated with structured models of use, including guided inquiry, POE or POEE-based approaches, critical prompting, reflective evaluation, and process-oriented assessment.

The thematic synthesis therefore suggested that generative AI did not automatically enhance higher-order thinking. Its

contribution depended on whether students were required to interrogate AI outputs, compare them with evidence, justify decisions, revise claims, and reflect on uncertainty. AI-supported learning appeared strongest when the tool was embedded within pedagogies that made validation, explanation, and metacognition explicit.

3.4.2.6 Theme 6. Institutional transition from prohibition to governed integration

The response patterns summarized in Table 3 demonstrated a consistent movement from restrictive approaches toward models of governed pedagogical integration.

The educational responses identified across the corpus included AI literacy, epistemic literacy, teacher training, curriculum integration, institutional policy, assessment redesign, prompt engineering, librarian-supported information literacy, and ethical AI guidelines. These responses suggested that higher education institutions had begun to move beyond prohibition or detection-based approaches.

The dominant response pattern was governed integration: AI was neither rejected nor adopted uncritically, but incorporated through policies, competencies, pedagogical models, and verification practices. This theme connected the empirical evidence with the broader need to prepare students and faculty for responsible participation in hybrid knowledge environments.

3.4.3 Analytical themes

Building upon the descriptive themes summarized in Table 3, the analytical synthesis generated five higher-order interpretive themes concerning the transformation of epistemic practices in AI-mediated higher education.

3.4.3.1 Analytical theme 1. From information access to epistemic responsibility

The synthesis indicated that generative AI had expanded access to explanations, texts, summaries, and academic support. However, this expansion also increased the burden of epistemic responsibility. Students were required not only to obtain information, but to evaluate its origin, reliability, coherence, evidentiary basis, and appropriateness for academic use. The core transformation was therefore not informational abundance, but the redistribution of responsibility for validating synthetic knowledge.

3.4.3.2 Analytical theme 2. From academic misconduct to hybrid authorship governance

The evidence suggested that the integrity debate had become insufficient when framed only through plagiarism or cheating. AI-mediated academic work required new norms for authorship, disclosure, accountability, and traceability. The analytical theme that emerged was hybrid authorship governance: the need to regulate and teach how human learners could ethically incorporate AI assistance while retaining intellectual ownership and responsibility.

3.4.3.3 Analytical theme 3. From digital competence to epistemic AI literacy

The reviewed studies repeatedly proposed AI literacy, but the synthesis showed that instrumental competence was not enough. Students needed to understand how AI systems generated plausible outputs, why those outputs could be wrong, how bias and hallucination could enter responses, and how claims should be verified. The required literacy was therefore epistemic AI literacy: the capacity to interrogate, validate, contextualize, and responsibly use AI-generated information.

3.4.3.4 Analytical theme 4. From cognitive assistance to calibrated delegation

Generative AI was frequently used to reduce cognitive load, accelerate tasks, and provide scaffolding. However, the synthesis showed that the educational problem was not delegation itself, but uncalibrated delegation. Productive AI use required knowing which cognitive and epistemic tasks could be supported by AI and which tasks needed to remain under human judgment, such as evaluating evidence, making final claims, interpreting uncertainty, and assuming responsibility for conclusions.

3.4.3.5 Analytical theme 5. From individual use to institutional epistemic governance

The corpus indicated that responsible AI use could not be left exclusively to individual students or teachers. Institutional governance was required through curriculum, assessment, faculty development, policies, library support, ethical guidelines, and competency frameworks. The analytical theme was therefore institutional epistemic governance: the creation of educational conditions that made AI-supported knowledge production transparent, accountable, and pedagogically meaningful.

3.4.4 Integrative thematic statement

Taken together, the descriptive and analytical patterns presented in Table 3 indicated a profound transformation in how knowledge was produced, validated, and governed within higher education environments increasingly mediated by generative AI systems.

Overall, the thematic synthesis indicated that generative AI had transformed higher education by introducing a hybrid epistemic environment in which students and faculty increasingly produced, refined, interpreted, and validated knowledge with the mediation of AI systems. The evidence suggested that the central educational challenge was not simply to prevent misuse, but to cultivate epistemic agency, calibrated trust, source evaluation, critical judgment, and responsible authorship. Generative AI was therefore interpreted as both a pedagogical scaffold and an epistemic risk amplifier. Its educational value depended on whether institutions and educators transformed AI use into an explicit object of critical, ethical, and evidence-based learning.

3.5 Framework synthesis

3.5.1 Framework synthesis procedure

Framework synthesis was conducted by mapping the thematic evidence generated in Thematic Synthesis onto the four-domain conceptual architecture developed for the review: epistemological, pedagogical, sociotechnical, and civic-democratic. The fit between the emergent thematic patterns and the predefined framework was assessed iteratively. When empirical findings aligned with one of the domains, they were retained as confirmatory evidence. When findings extended the scope of the original framework, the domain was refined through the addition of new interpretive components. This

TABLE 4 Framework synthesis matrix.

Framework Domain	Core Focus	Evidence Mapped from Thematic Synthesis	Confirmed Components	Extended or Refined Components	Contribution to HEEP-HE Model
Epistemological Domain	How knowledge was validated, justified, trusted, and attributed in AI-mediated academic environments	Reconfiguration of epistemic authority; fragility of knowledge validation; source evaluation; authorship; trust calibration; evidence checking	Authority, evidence, reliability, authorship, traceability, verification, justification	Synthetic plausibility, epistemic responsibility, hybrid authorship, calibrated trust, validation of AI-generated claims	Established the epistemic core of HEEP-HE: students and faculty were required to validate AI-mediated knowledge through verification, triangulation, justification, and responsibility for final claims
Pedagogical Domain	How students were taught to use, interrogate, and learn with generative AI	AI literacy; critical prompting; metacognitive instruction; inquiry scaffolding; assessment redesign; reflective use; guided experimentation	Critical thinking, AI literacy, information literacy, scientific literacy, authentic assessment, metacognition	Epistemic AI literacy, process-based assessment, prompt-mediated reasoning, pedagogy of verification, learner agency preservation	Defined the pedagogical mechanism of HEEP-HE: AI use became educationally valuable when embedded in structured tasks that required critique, reflection, evidence use, and metacognitive control
Sociotechnical Domain	How human learners and AI systems interacted within hybrid knowledge production ecologies	Cognitive offloading; overreliance; automation bias; hallucinations; algorithmic opacity; co-writing; AI-mediated feedback; delegation of cognitive tasks	Human–AI interaction, bias, opacity, hallucination, co-authorship, cognitive delegation, trust	Calibrated delegation, human oversight, synthetic information environments, AI as epistemic intermediary, distributed agency	Specified the sociotechnical condition of HEEP-HE: knowledge production became distributed across human and AI actors, requiring explicit governance of what could be delegated and what had to remain under human judgment
Civic-democratic Domain	How AI-mediated knowledge practices related to informed citizenship, misinformation, and democratic participation	Misinformation; public trust; evidence evaluation; scientific literacy; responsible use; institutional governance; ethical guidelines	Misinformation, scientific citizenship, evidence-informed reasoning, public trust, media literacy, democratic deliberation	Civic epistemic resilience, democratic knowledge validation, education against synthetic misinformation, institutional responsibility for epistemic citizenship	Expanded HEEP-HE beyond classroom practice: validating AI-mediated knowledge was interpreted as a democratic competence necessary for participation in synthetic information ecosystems

process allowed the synthesis to move from thematic aggregation toward conceptual model-building. (Table 4)

3.5.2 Domain-level interpretation

3.5.2.1 Epistemological domain

As shown in Table 3, the epistemological domain concentrated the strongest evidence generated by the synthesis. The included studies repeatedly indicated that generative AI had altered the conditions under which academic knowledge was produced, trusted, cited, and justified. The thematic findings concerning epistemic authority, hallucinated information, authorship, verification, and source evaluation confirmed the relevance of the original epistemological domain.

However, the evidence also required refinement of the initial framework. The traditional categories of evidence, reliability, authorship, and traceability were insufficient unless they were

expanded to include synthetic plausibility and epistemic responsibility. AI-generated outputs often appeared coherent and authoritative without necessarily being grounded in verifiable evidence. Therefore, knowledge validation in AI-mediated higher education required explicit practices of triangulation, claim-checking, citation verification, and human accountability for final academic assertions.

Within the HEEP-HE model, the epistemological domain therefore functioned as the core domain. It established that hybrid epistemic engagement involved not only using AI to access or produce information, but also actively validating AI-mediated claims through human judgment, evidentiary reasoning, and transparent attribution.

3.5.2.2 Pedagogical domain

As summarized in Table 3, the pedagogical domain captured how educational practices had responded to the epistemic

disruption introduced by generative AI. The thematic synthesis showed that AI literacy, critical prompting, metacognitive instruction, guided inquiry, and assessment redesign were recurrently proposed as educational responses. These findings confirmed the relevance of the original pedagogical domain, particularly its emphasis on critical thinking, information literacy, AI literacy, metacognition, and authentic assessment.

The framework was refined by distinguishing general AI literacy from epistemic AI literacy. The latter referred not only to knowing how to use generative AI tools, but to understanding how AI-generated outputs should be questioned, verified, contextualized, and responsibly incorporated into academic work. The evidence further suggested that effective AI integration required process-based assessment, reflective documentation of AI use, and pedagogical tasks that preserved learner agency rather than replacing student reasoning.

Within HEEP-HE, the pedagogical domain therefore operated as the instructional mechanism through which epistemic responsibility could be developed. Generative AI became educationally meaningful when its use was embedded in tasks requiring critique, comparison with evidence, justification of decisions, and reflection on uncertainty.

3.5.2.3 Sociotechnical domain

As detailed in Table 3, the sociotechnical domain addressed the interaction between human learners, institutional settings, and generative AI systems. The thematic evidence showed that AI tools had become active intermediaries in writing, feedback, explanation, synthesis, problem solving, and academic decision-making. This confirmed the relevance of framework categories such as human–AI interaction, bias, hallucination, opacity, cognitive delegation, and co-authorship.

Nevertheless, the evidence required the domain to be refined through the concept of calibrated delegation. The issue was not whether students used AI, but which epistemic tasks were delegated, under what conditions, and with what forms of oversight. Delegating brainstorming, language revision, or preliminary explanation differed from delegating final judgment, evidence validation, or interpretation of scientific claims. The sociotechnical dimension of the framework therefore had to account for distributed agency across humans and AI systems.

Within HEEP-HE, the sociotechnical domain specified the conditions of hybrid knowledge production. AI-mediated learning was interpreted as a distributed epistemic ecology in which students, teachers, institutions, and AI systems interacted. Responsible engagement required maintaining human oversight over evidence appraisal, ethical judgment, and final knowledge claims.

3.5.2.4 Civic-democratic domain

As presented in Table 3, the civic-democratic domain was less frequently addressed explicitly than the epistemological and pedagogical domains, but it remained conceptually central to the synthesis. The included studies repeatedly identified misinformation, unverified AI outputs, trust displacement, bias reproduction, and weakened critical judgment as risks. These patterns had implications beyond classroom performance because they concerned students' capacity to participate

in public knowledge environments shaped by synthetic information.

The framework was refined through the concept of civic epistemic resilience. This referred to the capacity to evaluate evidence, identify misinformation, interrogate algorithmically generated claims, and participate responsibly in democratic knowledge practices. The evidence suggested that validating AI-generated information was not merely an academic skill, but a civic competence linked to scientific literacy, media literacy, and evidence-informed participation.

Within HEEP-HE, the civic-democratic domain expanded the model beyond higher education classrooms. It positioned AI-mediated knowledge validation as part of the formation of scientifically informed citizens capable of navigating synthetic information ecosystems.

3.5.3 The HEEP-HE model: hybrid epistemic engagement practices in AI-mediated higher education

The HEEP-HE model is an evidence-informed conceptual product developed by the review team through framework synthesis. It was not directly proposed or empirically tested by any single included study and should not be interpreted as a validated intervention, measurement instrument, or causal model. Its components represent a review-level conceptual extrapolation grounded in recurrent patterns across the included evidence.

The framework synthesis led to the construction of the Hybrid Epistemic Engagement Practices in AI-Mediated Higher Education model. HEEP-HE was conceptualized as an integrative model explaining how students and faculty could engage responsibly with generative AI in higher education through practices that combined human judgment, AI-mediated support, evidentiary validation, pedagogical structure, sociotechnical awareness, and civic responsibility.

The model was organized around four interdependent domains:

1. Epistemological validation: the capacity to verify, justify, attribute, and take responsibility for AI-mediated knowledge claims.
2. Pedagogical scaffolding: the design of learning environments in which AI use supported critical thinking, metacognition, inquiry, and evidence-based reasoning.
3. Sociotechnical calibration: the ability to determine what could be delegated to AI, what required human oversight, and how algorithmic limitations shaped knowledge production.
4. Civic-democratic resilience: the development of competencies required to evaluate synthetic information, resist misinformation, and participate in evidence-informed democratic deliberation.

HEEP-HE therefore interpreted generative AI not as an external technological add-on to higher education, but as a mediator that reorganized epistemic practices. The model proposed that responsible AI-mediated learning depended on the alignment of

the four domains. Epistemological validation without pedagogical scaffolding risked remaining an abstract norm. Pedagogical innovation without sociotechnical calibration risked producing uncritical adoption. Sociotechnical awareness without civic-democratic orientation risked limiting the discussion to classroom efficiency. Civic-democratic aims without concrete validation practices risked remaining rhetorical.

The principal theoretical contribution of HEEP-HE was that it reframed generative AI use in higher education as a problem of hybrid epistemic engagement. Under this model, the central educational task was not to prevent AI use, nor to celebrate it as automatic innovation, but to cultivate learners capable of interrogating, validating, contextualizing, and ethically integrating AI-mediated knowledge. In this sense, HEEP-HE connected classroom practices of verification and authorship with broader democratic demands for scientific citizenship in synthetic information environments.

3.5.4 Refined framework derived from the evidence

The original four-domain framework was retained but refined through the synthesis. The epistemological domain was expanded through synthetic plausibility, epistemic responsibility, and hybrid authorship. The pedagogical domain was refined through epistemic AI literacy, pedagogy of verification, and process-based assessment. The sociotechnical domain was extended through calibrated delegation, distributed agency, and human oversight. The civic-democratic domain was strengthened through civic epistemic resilience and democratic validation of synthetic information.

Together, these refinements demonstrated that the evidence did not merely fit the initial framework; it extended it. The resulting HEEP-HE model is therefore presented as a provisional, evidence-informed conceptual contribution that requires independent empirical evaluation and validation.

4 Discussion

4.1 Interpretation of the empirical evidence

The principal empirical pattern identified by this review was not a uniform beneficial or detrimental effect of generative AI, but a context-dependent relationship between AI use and epistemic engagement. Across the included designs, verification-oriented and pedagogically structured uses of generative AI were associated with reflective engagement, conceptual clarification, iterative reasoning, and self-regulated learning. By contrast, convenience-oriented or weakly supervised uses were associated with overreliance, reduced source checking, superficial processing, and diminished evaluative engagement. This conditional interpretation is consistent with previous reviews emphasizing the importance of instructional scaffolding, active interrogation of AI outputs, and explicit verification practices (Li et al., 2026; Raitskaya and Tikhonova, 2025; Tomaszewski, 2025).

These patterns should nevertheless be interpreted cautiously. Much of the evidence was cross-sectional, exploratory, or based on participant perceptions and self-reported practices. The

review therefore does not establish that generative AI causes long-term cognitive decline, nor that pedagogical scaffolding independently produces sustained gains in critical autonomy. It identifies recurrent associations and experiential patterns within a recent and heterogeneous empirical literature.

4.2 Evidence-informed conceptual interpretation

On the basis of the framework synthesis, cognitive offloading can be interpreted as epistemic delegation when learners transfer not only task execution but also source selection, evidence appraisal, interpretation, verification, or final judgment to an AI system. This distinction is a review-level interpretation rather than an outcome directly measured in a standardized manner across the included studies. It extends existing concerns about AI dependency by identifying the specific epistemic operations that may be displaced when plausible AI-generated outputs are accepted without evidentiary scrutiny (Izquierdo-Condoy et al., 2025; Raitskaya and Tikhonova, 2025).

The HEEP-HE model was developed to organize this interpretation across four interdependent domains: epistemological validation, pedagogical scaffolding, sociotechnical calibration, and civic-democratic resilience. The model is not presented as an empirically validated intervention or causal theory. Rather, it is a provisional conceptual framework derived from recurrent findings concerning verification, authorship, trust calibration, learner agency, instructional design, and responsibility for synthetic information. Its principal contribution is to shift the analytical focus from whether students use AI to which epistemic functions they delegate, how those functions are supervised, and who retains responsibility for the resulting knowledge claims.

4.3 Normative educational and institutional implications

The following implications are normative propositions informed by, but not directly tested across, the included evidence. Higher education institutions should move beyond policies focused exclusively on prohibition, plagiarism detection, or disclosure of tool use. Responsible integration requires learning activities in which students compare AI-generated claims with external evidence, verify citations, identify uncertainty, explain acceptance or rejection decisions, and remain accountable for final interpretations.

Assessment practices should similarly emphasize process evidence rather than only the submitted product. Reflective records of AI use, source-verification logs, oral justification, iterative drafts, and explicit discussion of uncertainty may help make students' reasoning visible and reduce the substitution of AI output for independent judgment. These practices require accompanying faculty development, institutional guidance, access-sensitive implementation, and recognition that different disciplines have different evidentiary standards and acceptable forms of cognitive delegation.

4.4 Strengths and limitations

This review has several strengths. Study selection, data extraction, and methodological appraisal were conducted independently by two reviewers; the corpus included qualitative, quantitative, mixed-methods, quasi-experimental, and randomized designs; and the geographical distribution provided substantial representation from Global South higher education contexts. The combination of descriptive mapping, thematic synthesis, and framework synthesis enabled integration of a heterogeneous and rapidly developing body of evidence.

Several limitations must also be considered. First, the search was restricted to Scopus, the Web of Science Core Collection, and the databases searched through EBSCOhost. Relevant studies indexed exclusively in other education, technology, or regional databases may therefore have been missed. Second, grey literature was excluded, potentially increasing publication and availability bias and limiting the representation of institutional reports, dissertations, and unpublished evaluations. Third, the search ended on May 2, 2026; given the rapid rate of publication in this field, the review will require relatively early updating.

Fourth, the inclusion of early-access and online-first publications means that some bibliographic details may subsequently change, although publication status was verified against publisher and DOI records at the search cut-off. Fifth, conceptual and methodological heterogeneity precluded statistical pooling and limited direct comparison of outcomes. Sixth, a large proportion of the underlying evidence was cross-sectional, perception-based, or reliant on self-reported AI use, which restricts causal inference and may introduce recall and social-desirability bias. Seventh, the included studies operationalized critical autonomy, cognitive offloading, epistemic agency, and AI literacy inconsistently.

Finally, thematic and framework synthesis necessarily involve interpretive judgment. Although coding and theme development were conducted collaboratively and discrepancies were resolved through discussion, alternative analytical interpretations remain possible. The HEEP-HE model should consequently be regarded as a provisional evidence-informed framework rather than as an externally validated model.

4.5 Implications for future research

Future studies should employ longitudinal and experimental designs capable of measuring changes in epistemic performance over time. Greater use of objective tasks assessing citation verification, evidence discrimination, argument quality, uncertainty calibration, and independent transfer of learning would complement the current reliance on perceptions and self-reported behavior. Research should also examine discipline-specific forms of acceptable delegation, differential effects across student groups, and the influence of access, language, and institutional resources.

The HEEP-HE model itself requires empirical evaluation. Future research should examine whether its domains can be operationalized reliably, whether they predict meaningful

learning or epistemic outcomes, and whether educational interventions based on the model preserve learner agency while retaining the legitimate scaffolding benefits of generative AI.

5 Conclusion

The current empirical evidence indicates that generative AI can support feedback, conceptual clarification, reflective inquiry, and self-regulated learning when its use is structured around explanation, comparison, verification, and metacognitive reflection. The same technologies may contribute to overreliance, reduced source checking, cognitive offloading, and weakened evaluative engagement when they substitute for rather than support students' judgment. Given the predominance of recent, heterogeneous, and frequently self-reported evidence, these patterns should not be interpreted as proof of long-term causal effects.

The review further proposes HEEP-HE as a provisional evidence-informed framework for examining how epistemological validation, pedagogical scaffolding, sociotechnical calibration, and civic-democratic resilience interact in AI-mediated higher education. The model represents a conceptual extrapolation from the synthesis and requires independent empirical validation.

From a normative educational perspective, responsible integration should prioritize epistemic AI literacy, transparent authorship practices, process-based assessment, and explicit verification of AI-generated claims. Future longitudinal, experimental, and discipline-sensitive research is needed to determine how sustained generative AI use influences critical autonomy, epistemic agency, and the capacity to participate responsibly in synthetic information environments.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary Material](#), further inquiries can be directed to the corresponding author.

Author contributions

IS: Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing, Conceptualization, Funding acquisition, Supervision. MeS: Data curation, Formal analysis, Investigation, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. MaS: Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. AC: Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing, Investigation, Methodology, Project administration, Resources, Software.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. The authors used generative

artificial intelligence (AI) tools exclusively to support the linguistic revision and quality verification of the English language in the manuscript.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/feduc.2026.1878665/full#supplementary-material>

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